

BAN404 Exam V2025

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Load relevant packages for the exam

Task 1

Task 1 a:

Explain what the following R-functions are doing. Which method, treated in the course, is this a version of?

looks like a standard method, f is using difference in y and x multiplying by matrix b_1^2 and plussing with $la \cdot \text{sum of } b_1 \text{ squared}$. are we calculating error or deviation in the sample? check notes

g is also a function which takes the mean of the

linear discriminant analysis linear discriminant analysis assumes that the observations within each class (columns in the X matrix below) are drawn from a univariate normal distribution. This means that for the kth class

```

# X is an n x p matrix with data
# y is an n x 1 vector with data

f <-function(b1,X,y,la)
{
  fval <-sum((y-X%*%b1)^2)+la*sum(b1^2)
  return(fval)
}

g <-function(X,y,la)
{
  q <-ncol(X)   # store value for number of columns (k classes) to evaluate
  b0 <-mean(y)   # calculate mean of the dependent variable y
  yd <-y-b0      # calculate the error from the mean of the dependent variable y
  Xd <-X         # store matrix values in new matrix Xd

  for(k in 1:q) Xd[,k] <-X[,k]-mean(X[,k]) # Loop over all columns to xxx
  b1start<-rep(0,q) # create empty vector of size q to store values
  opt <-nlminb(b1start,f,X=Xd,y=yd,la=la) # optimize using port routines
  b1 <-opt$par    # store best set of parameters in b1
  return(b1)
}

```

Task 1 b

```

Xy <- read.csv2("data_task1.csv")
X <- as.matrix(Xy[,-1])
y <- as.matrix(Xy[,1])

# first method, using OLS

reg1b <- lm(y~X)

# second method, using la =1000
la1000 <- g(X,y,1000)

# using la=0
la0 <- g(X,y,0)

```

Task 1 c

```
# this function seems to evaluate the difference between observed error (diff between mean y
and yd (observed)) and the predicted using la (diff between ypred and yd observed)

loo <- function(la,X,y)
{
  n <- nrow(X) # store value of number of rows in X
  q <- ncol(X) # store value of number of columns in X
  b0 <- mean(y) # calculate mean of dependent variable y
  yd <- y- b0   # find error (difference from mean) for all rows in y
  Xd <- X       # create Xd matrix to input calculations below

  for(k in 1:q) Xd[,k] <- X[,k]- mean(X[,k]) # for all columns 1 to q the row position in col
umn subtracted the mean from that row is stored in Xd
                                     # meaning we store the error instead of the val
ue in the column.
  ydpred <- matrix(0,n,1)             # ydpred created and is a matrix of
  for(i in 1:n)
  {
    b1 <- g(X=X[-i,],y=y[-i],la)
    ydpred[i] <- Xd[i,]%*%b1 # times the best parameter b1
  }
  testMSE <- mean((yd-ydpred)^2)
  return(testMSE) # we can test different LA and see which minimizes testmse so we can find t
he optimal or close to optimal LA by plug and play?
}

MSEmin <- mat.or.vec(1,1000)
for (t in seq(from =0, to = 1000, by = 50))
  MSEmin[t] <- loo(t,X,y)

print(min(MSEmin)) # theoretically should find the minimum MSE of intervals 50 from 1 to 1000
on loo functiion but coding skills not able to write in R
```

```
## [1] 0
```

is it necessary to demean the x variables, why - why not? hmm i think the demean of the x is necessary (line 127) because it improves to evaluate the error when trying to

Task 1 d

Task 1 e

#Task 2

Task 2 a

Split the data into test and training data. Use descriptive statistics and graphs in order to find potentially useful predictors.

```
library(insuranceData)
data("data0hlsson")
ydata=data0hlsson
ydata$claim=ydata$antskad>=1

# remove variables that cannot be plausible as predictors

summary(ydata)
```

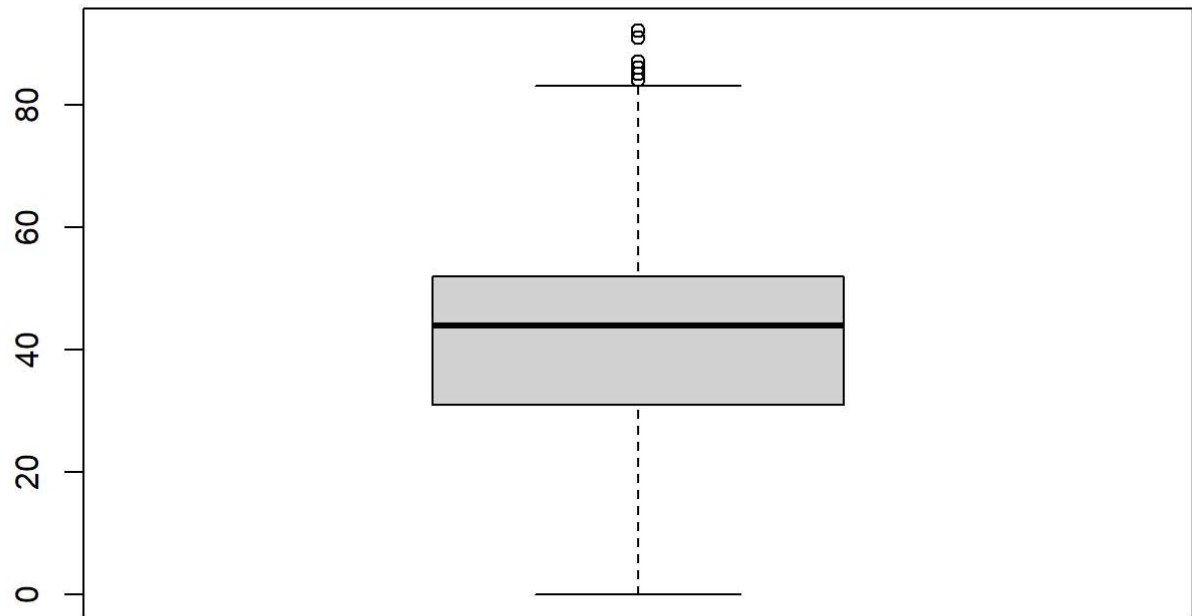
```
##      agarald      kon      zon      mcklass      fordald
## Min.   : 0.00    K: 9853   Min.   :1.000   Min.   :1.0   Min.   : 0.00
## 1st Qu.:31.00    M:54695   1st Qu.:2.000   1st Qu.:3.0   1st Qu.: 5.00
## Median :44.00                Median :3.000   Median :4.0   Median :12.00
## Mean   :42.42                Mean   :3.213   Mean   :3.7   Mean   :12.54
## 3rd Qu.:52.00                3rd Qu.:4.000   3rd Qu.:5.0   3rd Qu.:16.00
## Max.   :92.00                Max.   :7.000   Max.   :7.0   Max.   :99.00
##      bonuskl      duration      antskad      skadkost
## Min.   :1.000    Min.   : 0.0000    Min.   :0.0000    Min.   :      0
## 1st Qu.:2.000    1st Qu.: 0.4630    1st Qu.:0.0000    1st Qu.:      0
## Median :4.000    Median : 0.8274    Median :0.0000    Median :      0
## Mean   :4.025    Mean   : 1.0107    Mean   :0.0108    Mean   :    264
## 3rd Qu.:7.000    3rd Qu.: 1.0000    3rd Qu.:0.0000    3rd Qu.:      0
## Max.   :7.000    Max.   :31.3397    Max.   :2.0000    Max.   :365347
##      claim
## Mode :logical
## FALSE:63878
## TRUE :670
##
##
##
```

we observe that about 1 in a hundred policies has registered claims in the dataset. Males are highly overrepresented compared to women, and the median age of motorcyclists seem to be in the middle of their forties in (arguably in accordance with a common standard stereotype about the mid life crisis). from looking at this summary statistics it is difficult to immediately dismiss a variable, however it would be interesting to see if there is distinct differences between men and women, and also if vehicle type could be a good predictor or not, as well as age. We could however argue that variables such as skadcost (cost of a claim, is not a good predictor for claim) as it will only have values if a claim actually occurred, rendering it useless as a predictor because it inherently is behaving different for different situations we are measuring here. Antskad is also removed for this reason, but i want to note that it might be that someone already having a claim, might be more prone to behavior likely to cause outcomes in need of making a claim .

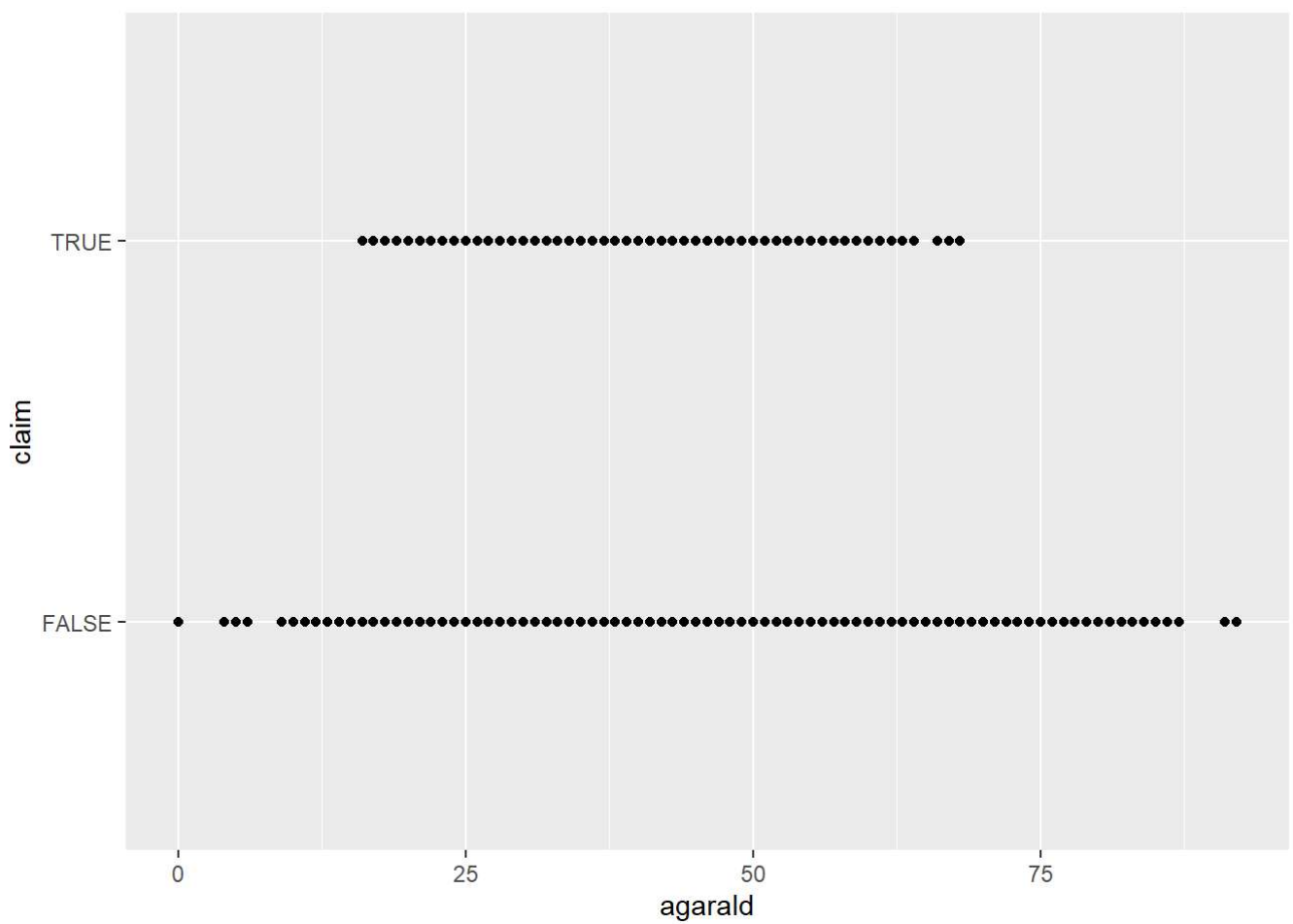
Examining age

one could imagine that, in general, age affects how our brain works, and thus the decision making of individuals. As the decisions made are often very important in determining outcomes in situations, it makes sense that age could be a predictor in determining claims. below we try and look a bit further into this.

```
boxplot(ydata$agarald)
```



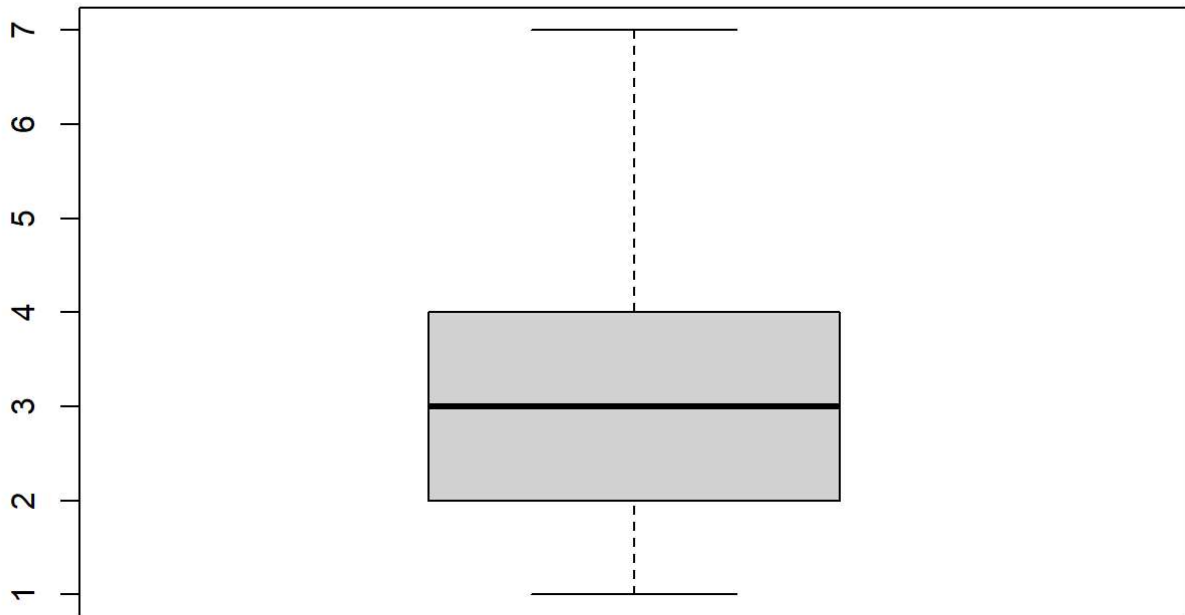
```
ggplot(data = ydata) +  
  geom_point(mapping = aes(x = agarald, y = claim))
```



Examining geographic zones

one could imagine that different geographic factors could make the road network more challenging to navigate, indicating it could have predictive quality in terms of claim. therefore looking a bit further into it before deciding to remove

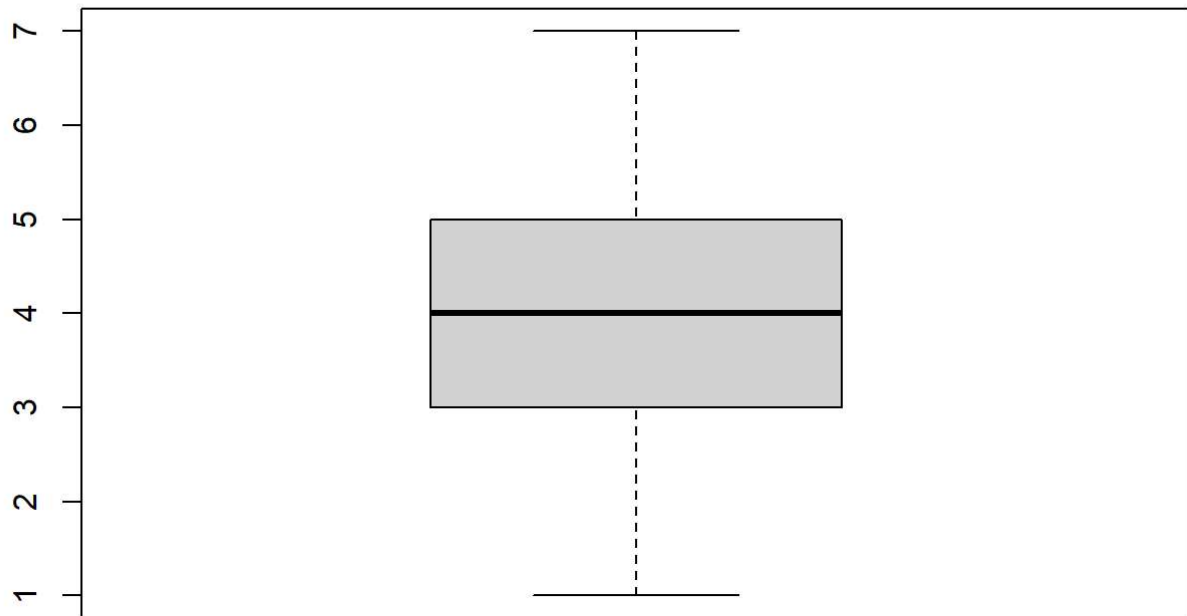
```
# trying to see about the data on areal zones  
boxplot(ydata$zon)
```



Examining vehicle class

Different types of vehicles behave differently, and one could argue that this in turn could possibly affect the rate of claims. below we therefore do a bit more research to identify possible relationship before removing.

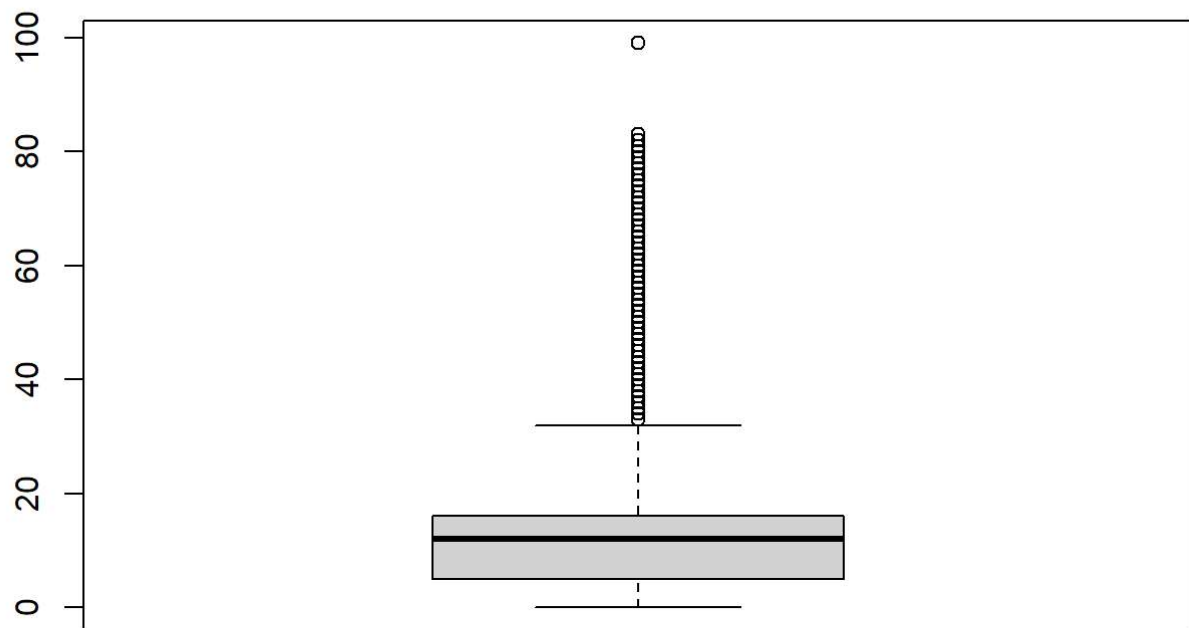
```
# trying to see if there is tendencies in vehicle class claims  
boxplot(ydata$mcklass)
```



Examining vehicle age

As mechanical objects age, wear and tear in turn make them inherently more likely to break down. we would therefore assume that the vehicle age is a good potential predictor for claims, and look a bit more into it below.

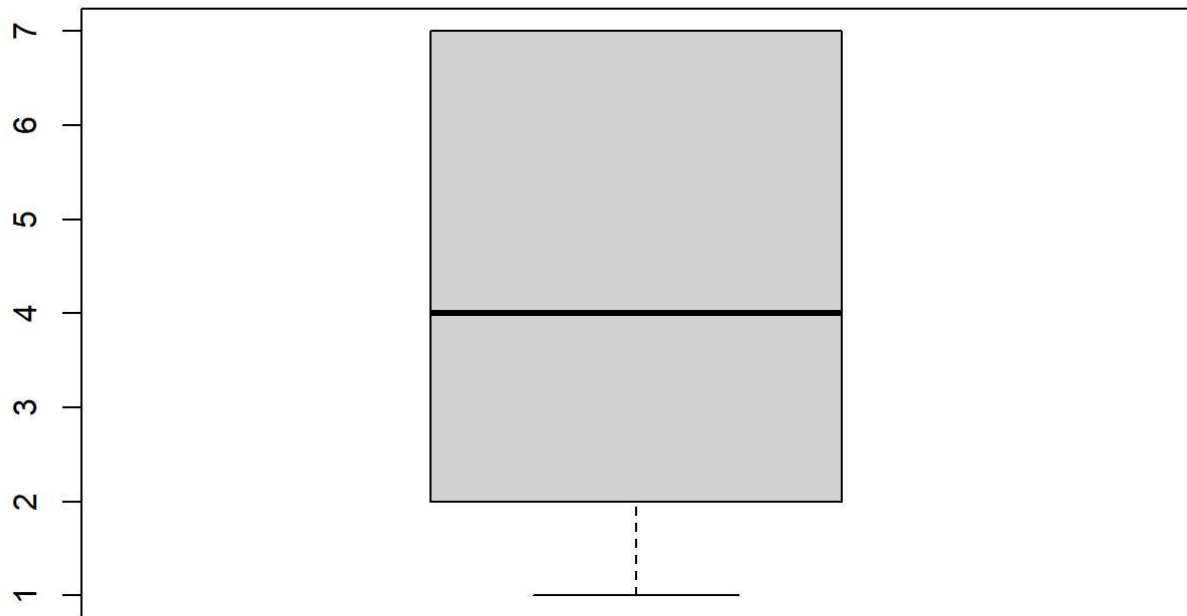
```
#  
boxplot(ydata$fordald)
```



Examining bonus class

bonus class is determined (at least in norway) among other factors, by the time since last a claim was made and one gets a “discount” because of this. In this way we would intuitively think that if a policy has a low bonus, they are either young and has not had time to build their bonus, or, a claim has been made in the near past (resetting the bonus). in such we would believe it could be a predictor, and will examine below if we can see a relationship using the eyeball test.

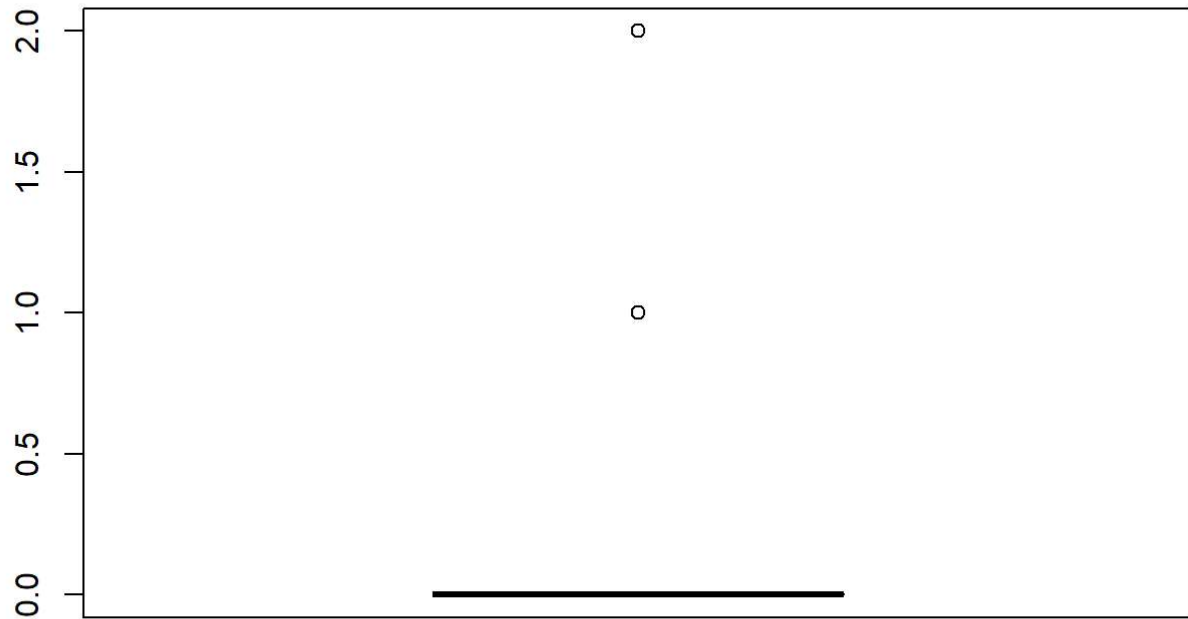
```
#  
boxplot(ydata$bonus1)
```

Examining antall skader

same reasoning as above, but will examine using a plot

```
boxplot(ydata$antiskad)
```



We have 2 classes that we are going to use in our logistic model, claim and not claim (claim = TRUE and claim = False)

```
# creating a logistic regression model
#yclaim <- ydata %>%
# if ydata$claim ==TRUE set value =1, if ydata$claim ==FALSE set value =0
#select(ydata$claim)

#ydata <- ydata %>%
# replace(ydata$claim,FALSE,"0")
```

Assuming we would have been able to replace False to 0, and True to 1, here is how we would have liked to go about fitting a logistic regression. First, we would split the data in half, test and train. Assuming randomized distribution patterns, train data could have contained the first half of the observations and test data the latter half of the data (32000 obs each circa). Then, given this is a logistic regression model, we would need to fit it in a probability setting, where the dependent variable has a probability of between 1 and 0. In this case, this would mean that $P(\text{claim} = 1 \text{ given } X = x \text{ predictors})$ is between 0 and 1 for all rows.

Task 2 b

Fit a logistic regression model with all variables and one with a selection of variables. The selection can be based on your arguments in task 2a and by the result of the model with all variables.

Task 2 c

Lets assume we have now fit a logistic regression model with all variable, i will call this "model1" and another logistic regression model only using the one predictor, lets call it "model2". We would expect model1 to be able better fitted to our train data in absolute terms, since it has a higher dimension of possible predicting variables.

However we would not be able to state beforehand that it would perform better in terms of AIC which takes into account the number of variables used to weight the models performance. In summary we would expect a higher r-squared from the larger model, but we would also be more worried about possible overfitting compared to the simpler model. Using the same modelling framework (multiple logistic regression), we would think intuitively that the best model would be located somewhere in the middle, where using the best predictors, that is not suffering from multicollinearity would make the best model to predict claims.

In terms of which threshold to choose: mathematically one could argue that a threshold of 0.5 would be logical since for all probabilities above this threshold, it would be more likely for a claim to happen (according to our model) than not. Based on how and where we are coming from though, one could argue the need to set the threshold higher than this, to make sure the predictive quality in a real world environment has a higher robustness. A model will inherently have limitations, and false positives will occur due to variability and chance in the real world. Another angle to look at it could also be as from an insurance provider standpoint, that we would aim to have a lower $p(\text{Claim})$ than 50% in order to overestimate the chances and therefore be able so secure that we anticipate possible claim prone customers in an overly fashion. It may cost more for the policies an insurance provider get wrong when not predicting claims than those customers they may lose because of a too high premium (wrongly predicted a claim when not happening). In summary, I argue this choice depends what we are after. If we want to reduce false negatives, we might want to lower our threshold below 0.5, if we want to minimize false positives, we might want to increase the threshold above 0.5.

Task 2 d

Fit a random forest and evaluate the predictions. Compare with the logistic model.

Lets assume that we have managed to fit the random forest model, also the multiple logistic model. Lets also assume that the model results is indicative of the general theory learned in this course. To make something rather than nothing, I will give a short discussion with this as the assumptions.

Given what we have learned about trees in this course, we know that trees generally have lower predictive accuracy than other regression and classifications approaches, and also that they can be non robust. However, when using ensemble methods such as random forest above, these weaknesses can be mitigated. If some predictors above was very strong and others were weak, we might overly focus on the strong predictor in our logistic model, as the effect from these overshadow weaker, yet maybe significant effects from the weaker predictors. This is something random forest fitting could mitigate since it forces each split to consider only a subset of the predictors when examining, decorrelating the trees. We might for example have a situation where several predictors above has multicollinearity (age, bonus, antalskad), and here, using a small m in our random forest could therefore improve the model if the scenario described is accurate.

Task 2 e

Fit a boosted tree and evaluate the predictions. Compare with the logistic model and the random forest.

Lets assume that we have managed to fit the random forest model, the boosted tree model and also the multiple logistic model. Lets also assume that the model results is indicative of the general theory learned in this course. To make something rather than nothing, I will give a short discussion with this as the assumptions.

Given what we have learned from the course regarding boosting in this context, we would think that the boosting approach could be a promising model for this dataset. As the boosting model does not fit a decision tree to the actual data, but rather the residuals, it learns more slowly, and performs better in areas where other model does not do well over time. Since the dataset we are working on is relatively big (compared to task 1), and we know that statistical models that learn slowly tend to perform well - we could argue that there is a likelihood of the boosted tree model to perform best out of the models in task 2.